

Bhrij Patel

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Education

University of Maryland, College Park

Ph.D. in Computer Science

Advisor: Dinesh Manocha

Research: Reinforcement Learning, Sequential Decision-Making, LLM Agents, Continual Learning, Representation Learning

College Park, MD

2022–Present

University of Maryland, College Park

M.S. in Computer Science

College Park, MD

May 2024

Duke University

B.S. in Computer Science & Mathematics, Minor in Creative Writing

Advisor: Cynthia Rudin

Durham, NC

May 2022

Research & Industry Experience

Emergence AI

Research Intern, Remote

Sept. 2026 – Present

Mentors: Aditya Vempaty, Ashish Jagmohan

- Designing a formal verification benchmark for LLM coding agents

GAMMA Lab, University of Maryland

Graduate Research Assistant

Aug. 2022 – Present

Mentors: Dinesh Manocha, Amrit Singh Bedi

- Developed a **Multi-Level Monte Carlo stochastic gradient estimator** for average-reward reinforcement learning, removing the need for oracle knowledge of Markov Chain mixing time while achieving $O(\sqrt{\tau_{\text{mix}}})$ dependence (**ICML 2023, 2024**).
- Designed **Confidence-Controlled Exploration (CCE)**, an uncertainty-guided exploration method for sparse-reward RL; improved navigation success by **18%**, reduced path length by **20–38%**, and reduced elevation cost by **9.3%** under a fixed training budget (**IROS 2025**).
- Developed **Code Comprehension then Auditing (CoCoA)**, a modular LLM evaluation framework that separates code comprehension from correctness judgment; improved F1 by up to **68%** and accuracy by up to **20%** over prior unsupervised evaluators across **9 datasets** and **6 LLM backbones** (*under review*).
- Developing a successor-representation approach decoupling environment learning from query-specific, graph retrieval, requiring **3–4× fewer training steps** than RL baselines to achieve comparable task completion rates (*in progress*).

Adobe Research

Research Intern, Bengaluru, India

Jun. 2026 – Sep. 2026

Mentors: Soumyabrata Pal, Ananya Sai, Ramasuri Narayanam

- Developed **Bench-Zero** to diagnose diverse agent failure modes and generate hard examples from limited to no data.
- Built an interactive demo that generates hard examples aligned with user-specified capability requests.

AG2 (formerly AutoGen)

Research Intern, Remote

Nov. 2025 – Feb. 2026

Mentor: Huazheng Wang

- Researched continual learning and feedback generation for tool-based, multi-turn LLM agents.
- Investigated failure modes of experience-based learning when stored trajectories contain noisy or misleading feedback.

Qualcomm Research — Efficient Agentic AI Team

Research Intern, Amsterdam, Netherlands

Jun. 2025 – Oct. 2025

Mentors: Davide Belli, Bence Major

- Developed **Dynamic Tool Dependency Retrieval (DTDR)**, modeling query semantics and execution history with lightweight predictors over frozen embeddings for resource-constrained LLM agents.
- Improved function-calling success rates by **23–104%** over SoTA static retrieval methods across **4 benchmarks and 7 LLM backbones (0.6B–14B parameters)** (**ACL 2026, Info Retrieval Workshop @ AAI 2026 - Oral**).
- Submitted 2 patent applications based on methods developed during the project.

Emergence AI

Research Intern, Remote

Feb. 2025 – May 2025

Mentors: Aditya Vempaty, Ashish Jagmohan

- Benchmarked methods for LLM agents to infer unknown API functionality from demonstrations and interaction experience **without API documentation**.
- Evaluated in-context learning and self-improvement methods across **3 agent benchmarks and 6 LLMs**, studying how agents infer tool functionality and parameter schemas from experience.

- Identified misleading execution feedback as a major failure mode, causing up to a **39% decrease in task success** (EMNLP 2025).

Selected Publications and Preprints

Code Comprehension then Auditing for Unsupervised LLM Evaluation

Bhrij Patel, Souradip Chakraborty, Mengdi Wang, Dinesh Manocha, Amrit Singh Bedi. arxiv.

Dynamic Tool Dependency Retrieval for Lightweight Function Calling

Bhrij Patel*, Davide Belli*, Amir Jalalirad, Maximilian Arnold, Aleksandr Ermolov, Bence Major. [ACL 2026](#).

Learning API Functionality from In-Context Demonstrations for Tool-based Agents

Bhrij Patel, Ashish Jagmohan, Aditya Vempaty. [EMNLP 2025](#).

Confidence-Controlled Exploration: Efficient Sparse-Reward Policy Learning for Robot Navigation

Bhrij Patel, Kasun Weerakoon, Wesley A. Suttle, Alec Koppel, Brian M. Sadler, Tianyi Zhou, Dinesh Manocha, Amrit Singh Bedi. [IROS 2025](#).

Towards Global Optimality for Practical Average Reward Reinforcement Learning without Mixing Time Oracles

Bhrij Patel, Wesley A. Suttle, Alec Koppel, Vaneet Aggarwal, Brian M. Sadler, Dinesh Manocha, Amrit Singh Bedi. [ICML 2024](#).

Beyond Exponentially Fast Mixing in Average-Reward Reinforcement Learning via Multi-Level Monte Carlo Actor-Critic

Wesley A. Suttle*, Amrit Singh Bedi*, **Bhrij Patel**, Brian M. Sadler, Alec Koppel, Dinesh Manocha. [ICML 2023](#).

* Equal contribution

Technical Skills

Programming: Python, Java

Machine Learning: PyTorch, Reinforcement Learning, Deep Learning, LLM Agents, Representation Learning, Embedding Models, Reward Modeling

Methods: Stochastic Optimization, Sequential Decision-Making, Monte Carlo Methods, Retrieval

Selected Awards

University of Maryland Dean's Fellowship (2022) | Duke DataFest Judges' Pick (2021) | COMAP Mathematical Contest in Modeling, Meritorious Winner (2020)